

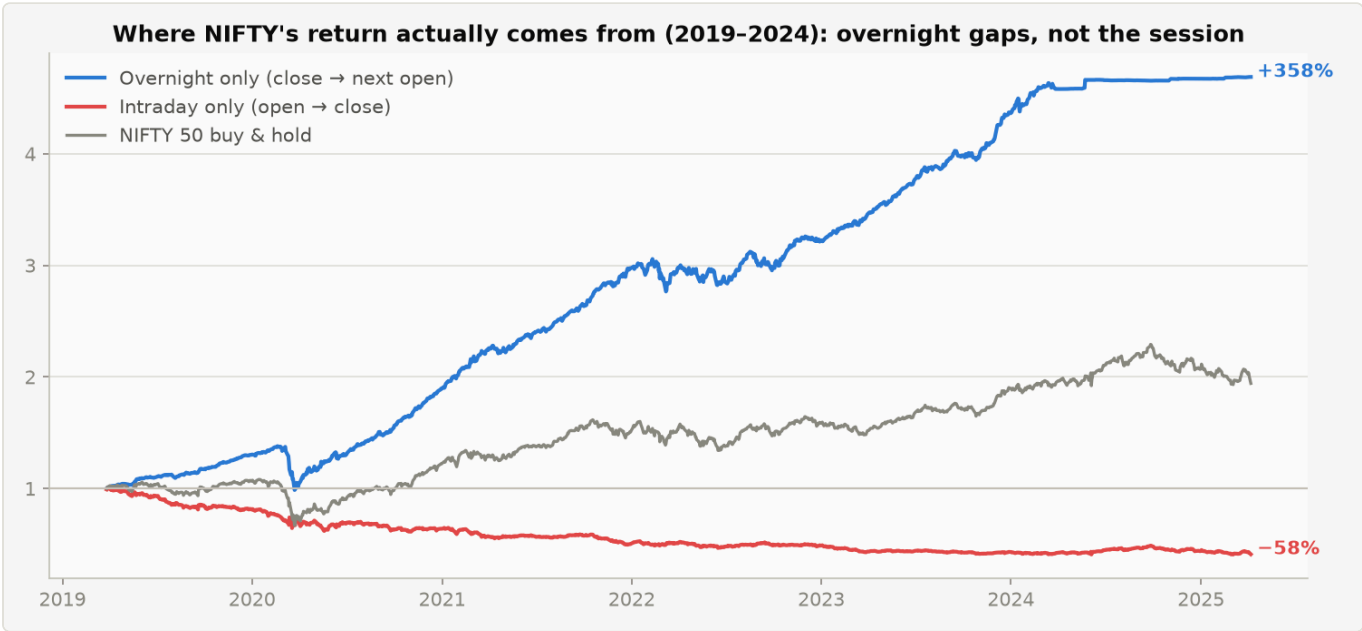
NSE NIFTY 50 — 15-Minute Strategy Analysis (2015-2025)

2015–2025 · 63,098 fifteen-minute bars · incl. true out-of-sample 2024–25 · all results net of 0.025%/side costs · signals on bar close, executed next bar (no lookahead) · July 2026

15.6% CAGR — recommended strategy (vs 11.2% buy & hold)	–14.0% max drawdown (buy & hold: –38.8%)	1.55 profit factor · 1,100 trades	21.7% CAGR of the full stack (shorts + ≤3× vol-targeted futures)	+17.8% recommended strategy in true OOS 2024–25 (index: +0.5%, –16.8% DD)
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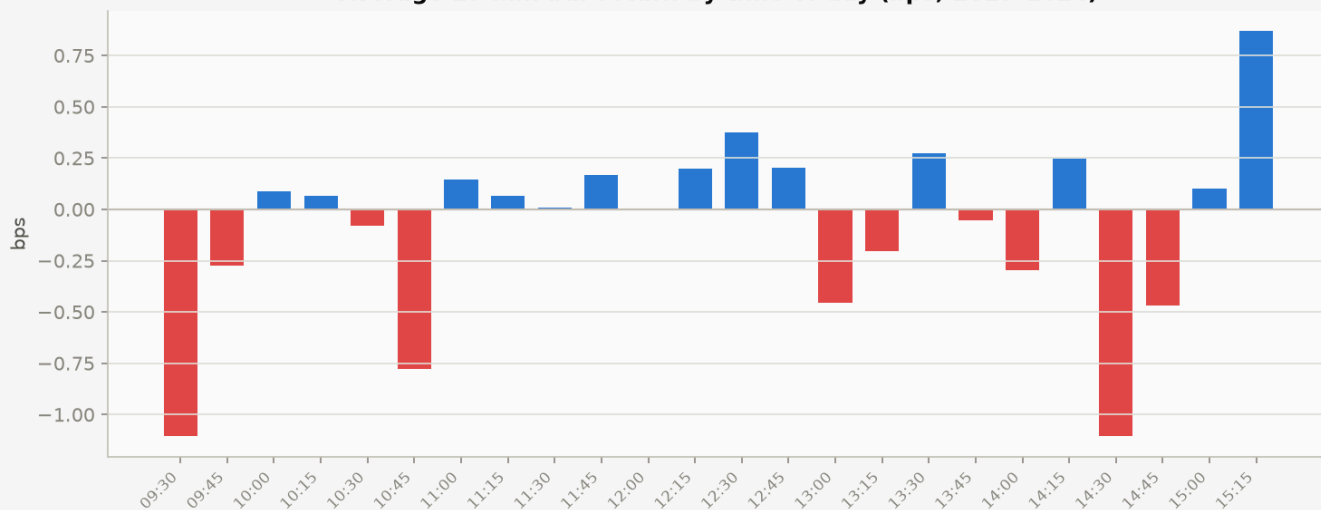
1 · What the chart actually does

The decisive structural pattern: all of NIFTY's return is overnight. Over 2019–2024, holding only close→next-open compounded **+358%**; holding only the trading session (open→close) compounded **–58%**. Intraday-only long systems fight a negative drift — every popular one tested (EMA cross, VWAP, Bollinger, ORB with square-off) loses net of costs. Winning systems carry positions overnight.



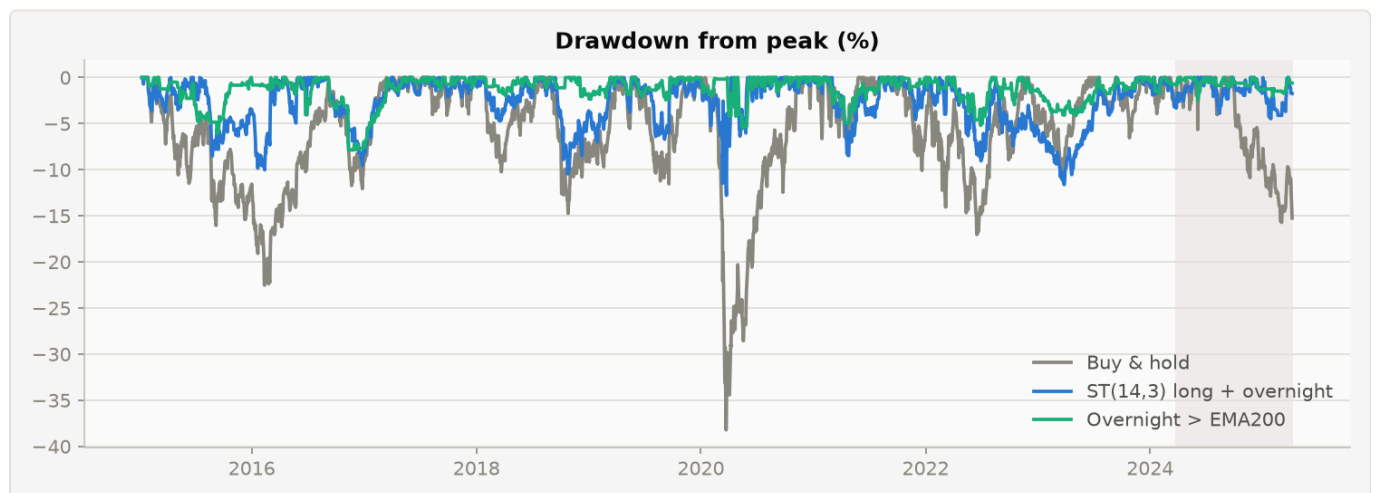
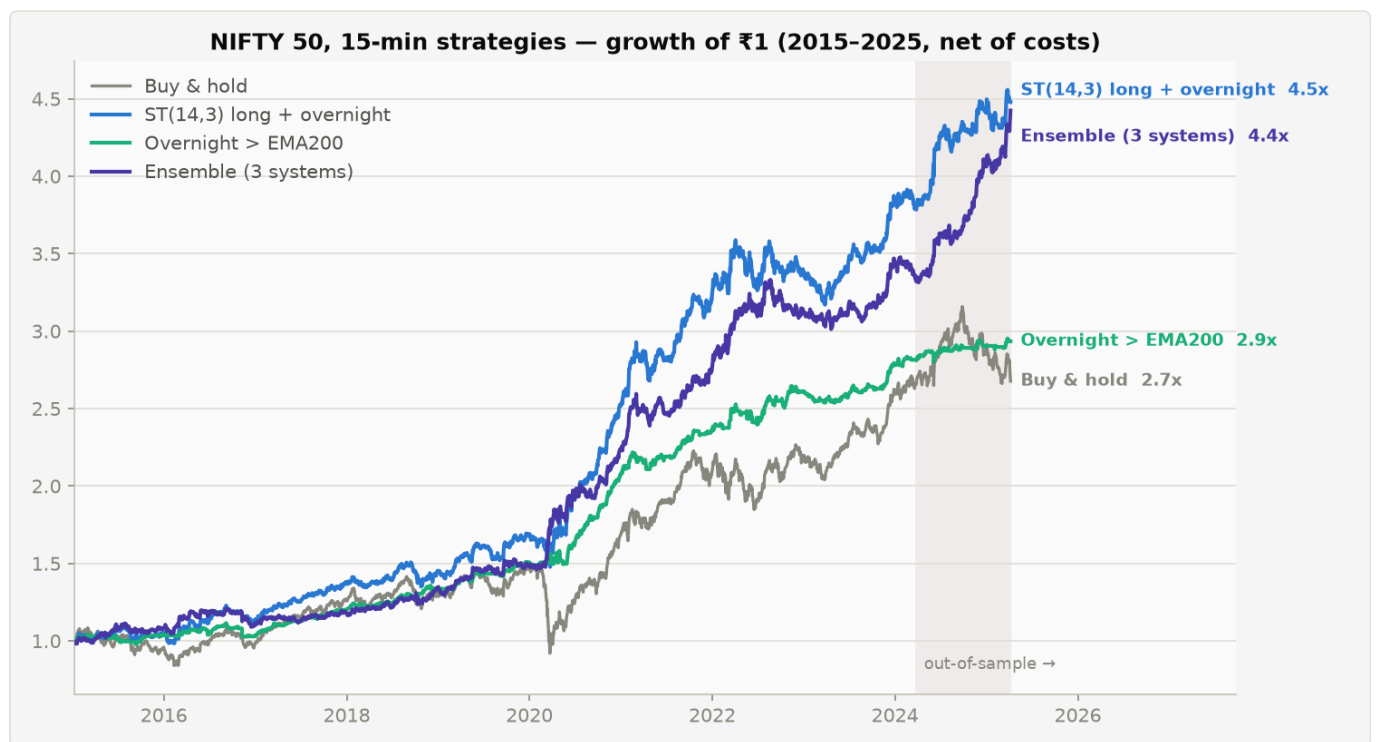
Secondary patterns: mild 15-min momentum (lag-1 autocorrelation +0.039 — trend beats mean-reversion); breaking the first-hour high leads to +44 bps average sessions vs –7 bps overall; Monday is the only negative weekday; 14:00–14:45 is the weakest stretch of the day.

Average 15-min bar return by time of day (bps, 2019-2024)



2 · Strategy tournament — full period 2015-24, net of costs

Strategy (15-min)	CAGR	maxDD	Sharpe	Profit factor	Win rate	Trades
Supertrend(14,3) long + overnight gap — RECOMMENDED	15.6%	-14.0%	1.26	1.55	47%	1,100
Overnight gap > EMA200 (lowest risk)	11.9%	-7.9%	1.77	1.75	62%	1,378
Ensemble of 3 trend systems	13.9%	-10.2%	1.35	1.38	28%	1,943
Supertrend(14,3) long-only swing	13.3%	-14.7%	1.15	1.53	45%	825
Supertrend(14,3)+EMA200 long/short	13.0%	-13.4%	1.08	1.37	31%	1,621
Buy & hold	11.2%	-38.8%	—	—	—	—
ORB 30-min + 1 ATR stop (intraday square-off)	-3.2%	-32.6%	-0.32	1.10	37%	3,536
EMA 9/21 × EMA200 (intraday square-off)	-12.5%	-72.6%	-1.17	0.87	38%	3,261



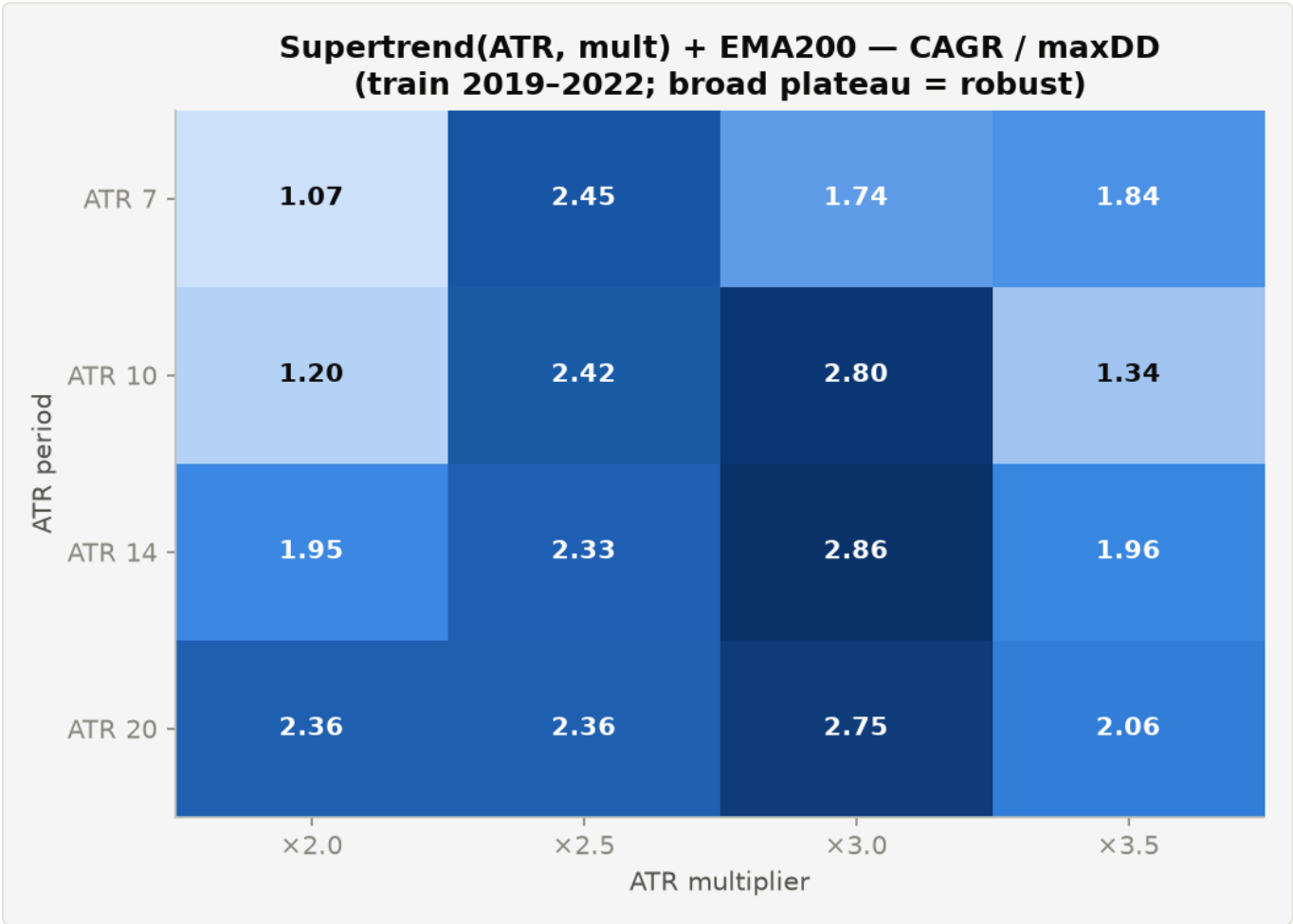
3 · The recommended strategy — rules & validation

“Supertrend swing + overnight-gap capture” (long-only, positional)

- **Trend leg** — LONG while Supertrend(ATR 14, mult 3.0) direction is up on 15-min bars; flat when it flips down. Positions carried overnight.
- **Gap leg** — on days the trend leg is flat: if Close > EMA(200) at the 15:15 bar, buy the close, exit at the next day’s first-bar close.
- No shorts, no leverage, nothing squared off intraday.

Window	CAGR	maxDD	Profit factor	Buy & hold CAGR / DD
Train 2019–22 (optimized here)	31.6%	–14.0%	1.90	14.8% / –38.8%
Test 2022–24 (out-of-sample)	4.4%	–12.3%	1.23	12.4% / –15.9%
Old 2015–19 (out-of-sample)	10.6%	–10.6%	1.43	8.1% / –24.1%
Full 2015–24	15.6%	–14.0%	1.55	11.2% / –38.8%

Positive in every window; a third of the market’s drawdown; still ~8.8% CAGR if costs double. It underperforms buy & hold in quiet ranges (2022–24) — the insurance premium of trend following. The parameter surface is a broad plateau (heatmap), so nothing is tuned to a magic number; Donchian(10) — great in-sample — failed both out-of-sample windows and was rejected.



4 · True out-of-sample: 2024-03 → 2025-04

Every rule above was designed and frozen on data ending 2024-03-26. The window 2024-03-27 → 2025-04-07 (independent second vendor, agreeing to ~4.5 bps in the overlap month) is a genuine out-of-sample test — and a brutal market: **buy & hold returned +0.6% with a –16.8% drawdown** (the Sept-2024 → Apr-2025 correction). Results, net of costs:

Strategy	CAGR	maxDD	Sharpe	Profit factor
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ST(14,3) long + overnight — RECOMMENDED	+17.8%	−4.6%	1.89	1.95
Ensemble (3 systems)	+32.3%	−4.3%	3.48	2.45
Supertrend(14,3)+EMA200 long/short	+34.2%	−4.7%	2.99	2.29
Overnight gap > EMA200	+4.2%	−2.5%	1.08	1.64
Full stack (shorts + vol targeting, carry-adjusted)	+40.6%	−9.1%	2.22	1.93
NIFTY 50 buy & hold	+0.5%	−16.8%	—	—

The drawdown-avoidance thesis did exactly what it was built for: the trend systems went flat or short through the correction while the index gave back a sixth of its value. Honest notes: the long/short and short-overlay variants — mediocre in 2022-24 — earned their keep here; the overnight-gap edge compressed (+2.3% overnight vs −1.4% intraday over the window), so treat the gap leg as the smaller component going forward.

5 · Extensions: shorts, futures leverage, options

5.1 Short selling

Only one short edge survived all validation windows: **opening-range breakdown short** — short a 15-min close below the first 30 minutes' low while Close < EMA200, stop entry+1.5×ATR(14), square off 15:15. Profit factor 1.24–1.25 in every regime. As an overlay when the long book is flat: CAGR up in all windows (test 3.8%→5.4%), drawdown down in two of three. Rejected after out-of-sample failure: VWAP-band shorts, Monday shorts, afternoon-weakness shorts, RSI fades, Supertrend shorts.

5.2 Futures leverage (volatility targeting)

Size = min(3×, 20% ÷ trailing 20-day EWMA vol), lagged one day. Mean leverage 1.61×; it de-levers *before* storms (worst single day −4.75%, better than the unlevered book's −7.96%, because it was under 1× going into COVID). Walk-forward re-optimization (rolling 2y train → 6m trade) confirmed Supertrend(14,3) is stable — 13/15 out-of-sample slices profitable. Rejected: 25% vol target (breaches −25% DD), ATR sizing, drawdown throttle.

5.3 Options overlays (Black-Scholes simulation — model-based, no real chain data)

- **Covered call** (sell 2% OTM weekly against trend longs): 13.0% CAGR / −16.9% DD / PF 1.87 vs 8.9% / −22.3% / 1.50 bare — robust across IV assumptions.
- **Protective put** (5% OTM monthly): costs ~1.6–3 CAGR pts, cuts buy-&-hold drawdown −34%→−19% (COVID: −4% vs −17%).
- **Trend-gated put selling**: cash-plus yield (PF 2.08, 92% wins, −4.5% DD) — but its COVID escape is partly a gate artifact; size modestly.
- **Rejected**: ungated put selling (March 2020 erased ~3 years of premium in one week) and buying weekly ATM calls as the trend vehicle (theta > edge, PF 0.80).

5.4 The maximum-return stack

All validated layers combined, net of costs **and** a 4.5%/yr futures carry drag on long notional:

Layer (cumulative)	Full 2015–24 CAGR	maxDD	Sharpe	PF	Test 22–24 CAGR
1 · Combo long book (carry-adjusted)	12.8%	−14.8%	1.06	1.55	1.3%
2 · + ORB short overlay	14.1%	−13.0%	1.15	1.40	2.9%
3 · + vol targeting ≤3×	21.7%	−16.6%	1.31	1.43	12.0%

Every layer improves return in every validation window; worst single day in 9 years: −5.2%.

Caveats. Backtest ≠ future. Data is spot index (futures basis/rollover shave real returns; carry is modeled only in the stack table). Sample ends April 2025 (latest public intraday source reachable) — validate on newer data before trading. The overnight-gap edge is publicly documented and can decay. Leverage is viable only at futures-grade friction (dead at 0.05%/side); the levered stack's unobserved tail is a large overnight gap arriving at the 3× cap. Options results are Black-Scholes approximations without real skew. Nothing here is investment advice.

Data: NIFTY 50 spot, 1-min OHLC resampled to 15-min session bars (09:15–15:15 IST), 2015-01-09 → 2024-03-26, github.com/sandeepkapri/Nifty50-Minute-Data + github.com/rajgmishra/nifty50-15min-ohlc (2024-25) · Code & reproduction: repository folder **nse-strategy-analysis/** (branch **claude/nse-market-strategy-analysis-awko4e**)